

6. A group of students use 6-sided dice to model radioactive decay.
There are 240 dice in total.
Dice with a 6 facing upward represent decayed nuclei.



- (a) (i) State what the students should do after throwing the 240 dice for the first time. [1]

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- (ii) Describe how they should continue the experiment. [1]

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- (iii) I. State the chance (probability) of any one of the dice landing with a 6 facing upward. [1]

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- II. Use your previous answer to predict how many dice will land with a 6 facing up on the first throw. [1]

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(b) The table shows the results.

Throw	Number of sixes	Number of dice remaining
0	0	240
1	38	202
2	30	
3	28	144
4	22	122
5	22	100

- (i) **Complete** the table.

[1]
- (ii) State how many dice remain after one half-life.

[1]
- (iii) Use the data in the table to estimate the half-life of the ‘dice’ in throws.

[1]

Half-life = throws

7

